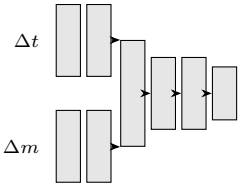
1. Adversarial Examples Generation

1.1 Surrogate two-tower CNN training



1.2 Photometrically bounded PGD attack

- Observed light curve $\{(t_i, m_i, e_i)\}$
- PGD on the Δm channel (T=20 steps, smoothness penalty λ_{sm}=10^{-3})
- Final hard projection in raw magnitudes: $m_i^{\text{adv}} \leftarrow \text{clip}\big(m_i^{\text{adv}},\ m_i - k\,e_i,\ m_i + k\,e_i\big)$
- Pointwise budget $|m_i^{\text{adv}} - m_i| \le k\,e_i$, k=0.5
- Adversarial light curve saved to disk

1.3 Transferability property

$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\text{RF}} P(\hat{y} \mid x_{\text{adv}})$$
$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\text{CNN}} P(\hat{y} \mid x_{\text{adv}})$$
$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\text{RNN}} P(\hat{y} \mid x_{\text{adv}})$$

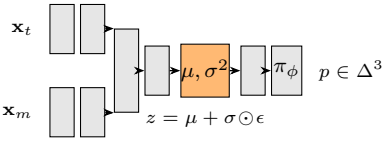
adv. light curves

2. Robust Raw Encoder (RRE) with adversarial training

2.1 Raw differential representation (no period, no folding)

- $\Delta t_i = t_i - t_{i-1}, \quad \Delta m_i = m_i - m_{i-1}$
- $\sigma_{\Delta m_i} = \sqrt{e_i^2 + e_{i-1}^2}$ (propagated uncertainty)
- Uniformly subsample/pad to $L = 500$ epochs; global per-channel standardization (train-split statistics)
- Two-tower input: $\mathbf{x}_t = \Delta t \in \mathbb{R}^{L\times 1}$, $\mathbf{x}_m = [\Delta m, \sigma_{\Delta m}] \in \mathbb{R}^{L\times 2}$

2.2 Two-tower encoder with variational bottleneck



Per-channel Conv1D towers + joint conv block + GAP, latent $D = 48$

2.3 Inner PGD attack in raw-magnitude space

$\epsilon_i \leftarrow k\,e_i \quad \delta \sim \mathcal{U}(-\epsilon, \epsilon)$, δ_1=0 (anchor)
for $s = 1, \ldots, S = 3$ **do**
  $\mathbf{x}' \leftarrow$ recompute Δm from m+δ (differencing)
  $\delta \leftarrow \delta + 0.4\,\epsilon\ \text{sign}\,\nabla_\delta\,\text{CE}\Big(y,\ \pi_\phi(\mu_\theta(\mathbf{x}'))\Big)$
  $\delta \leftarrow \text{clip}(\delta, -\epsilon, +\epsilon) \quad \Rightarrow |m_i^{\text{adv}} - m_i| \le k e_i$ always
return $\mathbf{x}_{\text{adv}}$

2.4 Combined training objective

$$\mathcal{L} = \underbrace{\text{CE}(y,p)}_{\text{clean}} + \lambda_{\text{adv}} \underbrace{\text{CE}(y, p_{\text{adv}})}_{\text{adv-CE}} + \lambda_{\text{tr}} \underbrace{\text{KL}(p \,\|\, p_{\text{adv}})}_{\text{TRADES}} + \lambda_{\text{sc}} \underbrace{\mathcal{L}_{\text{SupCon}}(z, z_{\text{adv}}, y)}_{\text{latent consistency}} + \beta \underbrace{\text{KL}\Big(q_\theta(z|x) \,\|\, \mathcal{N}(0, I)\Big)}_{\text{VIB compression}}$$

one shared $\mathbf{x}_{\text{adv}}$ feeds all adversarial terms;
$(\lambda_{\text{adv}}, \lambda_{\text{tr}}, \lambda_{\text{sc}}, \beta) = (1,\ 2,\ 0.5,\ 5 \times 10^{-4})$

2.5 Robust classifier evaluation

$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\text{RF}} P(\hat{y} \mid x_{\text{adv}})$$
$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\text{CNN}} P(\hat{y} \mid x_{\text{adv}})$$
$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\text{RNN}} P(\hat{y} \mid x_{\text{adv}})$$
$$\{x_{\text{adv}}\} \in \mathbb{R}^n \xrightarrow{\textbf{ours (RRE)}} P(\hat{y} \mid x_{\text{adv}})$$

+ adaptive white-box PGD on RRE itself
(raw-magnitude budget, 40 iter., 5 random restarts)